

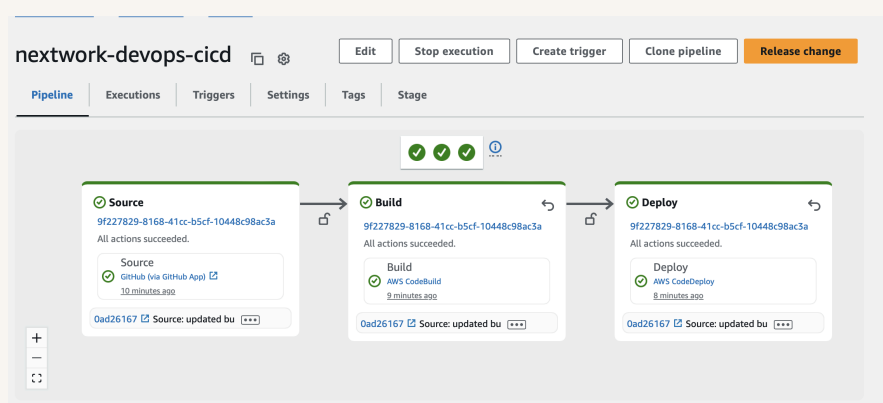


nextwork.org

Build a CI/CD Pipeline with AWS



delorezdoris@gmail.com





Introducing Today's Project!

In this project, I will demonstrate how to create a complete CI/CD pipeline with AWS CodePipeline. I'm doing this project to show how to connect your source code, build, and deployment stages. It will also show how to test automatic deployments.

Key tools and concepts

Services I used were code pipeline, code deploy, code artifact, cloud formation, IAM. Key concepts I learnt include the different stages in the CI/CD pipeline, handling rollbacks and webhooks.

Project reflection

This project took me approximately 3 hours. The most challenging part was having to renew my git hub token. It was most rewarding to see my live deployed webapp update without having to build and deploy the project myself.

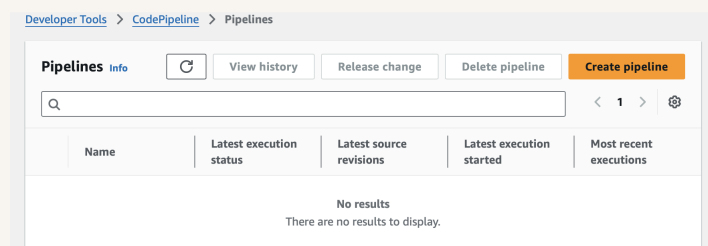


Starting a CI/CD Pipeline

AWS CodePipeline is a CI/CD tool on which you can create a workflow that automatically moves your code changes through the build and deployment stage. Using CodePipeline makes sure your deployments are consistent, reliable and happen automatically.

CodePipeline offers different execution modes based on how the executions are processed. I chose Superseded mode, if a new pipeline execution is triggered while another execution is already in progress, the newer execution will take over.

A service role gets created automatically during setup so code pipeline has permission to access other AWS resources it needs to run our pipeline, such as S3 buckets for storing artifacts or CodeBuild for building the code.

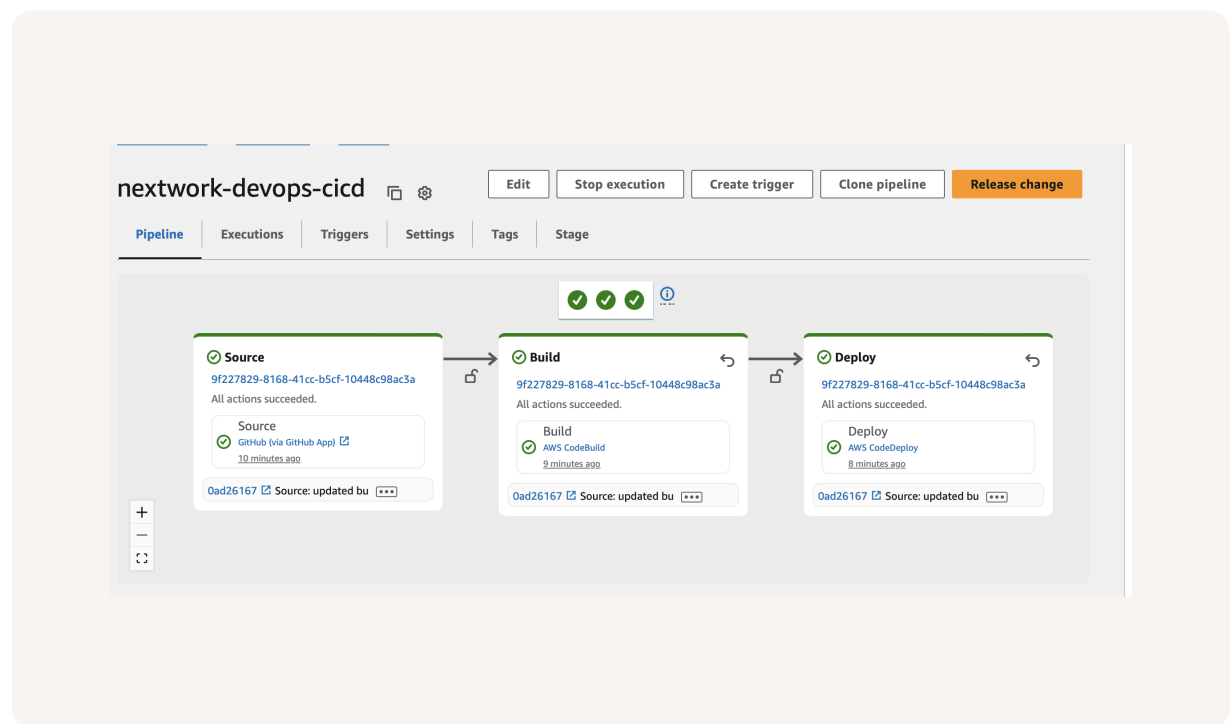




CI/CD Stages

The three stages I've set up in my CI/CD pipeline are the source stage, build stage and the deploy stage. While setting up each part, I learnt about the various execution modes.

CodePipeline organizes the three stages into a single diagram that showcases the flow of changes from stage to deploy. In each stage, you can see more details on the pipeline execution that it belongs to and the shortcuts to the connected service.





Source Stage

In the Source stage, the default branch tells CodePipeline to monitor this branch for changes and trigger the pipeline whenever there's a commit to it.

The source stage is also where you enable webhook events, which lets CodePipeline automatically start your pipeline whenever code is pushed to your specified branch in GitHub.

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via GitHub App) ▼

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

arn:aws:codeconnections:us-east-1:699475956117:connection/161c X ↻ or [Connect to GitHub](#)

Repository name
Choose a repository in your GitHub account.

Doris-01/nextwork-web-project X

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.

Default branch
Default branch will be used only when pipeline execution starts from a different source or manually started.

main X

Output artifact format
Choose the output artifact format.

☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.

☐ **Full clone**
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions. [Learn more](#)

☒ **Enable automatic retry on stage failure**



Build Stage

The Build stage sets up where your source code gets compiled and packaged .The input artifact for the build stage is the source artifact which is the ZIP file containing our source code that was outputted by the Source stage.

Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

nextwork-devops-cicd

or

Create project

☐ Define buildspec override - *optional*
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

Environment variables - *optional*
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

Add environment variable

Build type

☒ Single build
Triggers a single build.

☐ Batch build
Triggers multiple builds as a single execution.

Region

United States (N. Virginia)



Deploy Stage

The Deploy stage is where application artifacts (the output from the Build stage) are deployed to the target environment .

Build provider
Choose the tool you want to use to run build commands and specify artifacts for your build action.

☐ Commands

☒ Other build providers

AWS CodeBuild

Project name
Choose a build project that you have already created in the AWS CodeBuild console. Or create a build project in the AWS CodeBuild console and then return to this task.

nextwork-devops-cicd

or

Create project

☐ **Define buildspec override - optional**
Buildspec file or definition that overrides the latest one defined in the build project, for this build only.

Environment variables - optional
Choose the key, value, and type for your CodeBuild environment variables. In the value field, you can reference variables generated by CodePipeline. [Learn more](#)

Add environment variable

Build type

☒ **Single build**
Triggers a single build.

☐ **Batch build**
Triggers multiple builds as a single execution.

Region
United States (N. Virginia)



Success!

Since my CI/CD pipeline gets triggered by code changes and webhook events. I tested my pipeline by updating my code. This means I added a new line to the webapps index.jsp file and pushed those changes.

The moment I pushed the code change code pipeline responded by triggering a new build and deploy stage. The commit message under each stage reflects the latest code change that is being used.

Once my pipeline executed successfully, I checked my live webapp and confirmed that it updated without having to manually rebuild and redeploy our code. The changes went straight to the production environment.



Hello Doris!

If you see this line in Github, that means your latest changes are getting pushed to your cloud repo :o

If you see this line, that means your latest changes are automatically deployed into production by CodePipeline!



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